

CLEVELAND BURKE LAKEFRONT, OH, USA

WMO: 725245

Lat: 41.518N

Lon: 81.684W

Elev: 178

StdP: 99.20

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 04853

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.7	-11.1	-19.1	0.7	-12.9	-16.5	0.9	-10.1	15.0	-2.3	13.9	-4.4	6.0	220	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.6	31.5	23.0	30.0	22.6	28.6	22.2	24.6	29.0	23.9	28.1	23.2	27.3	5.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.1	27.2	22.5	17.5	26.9	22.0	17.0	26.5	75.3	29.2	72.4	28.1	69.6	27.5	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.7	11.3	10.1	DB	-15.9	34.5	3.8	1.3	-18.6	35.5	-20.8	36.2	-23.0	37.0	-25.7	38.0
			WB	-16.6	26.0	3.7	1.0	-19.2	26.7	-21.4	27.3	-23.5	27.9	-26.1	28.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	-1.3	-0.3	3.5	10.0	16.3	21.5	23.7	23.1	19.8	13.5	7.7	1.9	(6)
(7)		DBStd	10.11	6.13	5.95	5.90	5.09	4.78	3.79	2.71	2.55	3.71	4.50	4.75	5.36	(7)
(8)		HDD10.0	1305	354	293	219	62	3	0	0	0	16	100	259		(8)
(9)		HDD18.3	2978	609	522	461	256	99	14	1	26	161	318	511		(9)
(10)		CDD10.0	1917	3	3	17	61	199	344	426	407	295	124	32	6	(10)
(11)		CDD18.3	552	0	0	1	5	37	108	168	150	71	12	1	0	(11)
(12)		CDH23.3	3592	0	0	4	51	241	716	1224	904	390	60	1	0	(12)
(13)		CDH26.7	867	0	0	0	7	57	208	306	189	94	7	0	0	(13)
(14)	Wind	WSAvg	4.9	6.1	5.5	5.2	5.0	4.2	3.9	3.7	3.8	4.2	5.2	5.6	5.9	(14)
(15)	Precipitation	PrecAvg	1023	76	58	71	89	95	97	106	90	100	88	73	80	(15)
(16)		PrecMax	1334	154	139	153	147	186	202	235	199	228	172	127	180	(16)
(17)		PrecMin	716	37	24	16	44	35	21	32	32	30	37	31	40	(17)
(18)		PrecStd	145	29	25	28	28	44	42	48	35	44	31	26	28	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.4	19.9	23.8	27.6	30.5	33.0	33.2	32.9	32.1	27.4	22.2	18.4	(19)
(20)			MCWB	13.6	13.7	15.4	17.4	21.3	22.8	24.0	23.1	22.2	19.9	15.7	14.0	(20)
(21)		2%	DB	13.8	14.7	19.9	24.7	28.2	31.1	31.2	30.4	29.1	25.0	19.0	15.0	(21)
(22)			MCWB	11.0	10.8	13.5	16.5	20.6	22.5	23.7	23.1	21.9	18.5	14.0	11.6	(22)
(23)		5%	DB	10.6	11.1	16.9	21.9	26.2	28.9	29.4	28.6	27.3	22.7	17.3	12.2	(23)
(24)			MCWB	7.5	7.4	11.8	14.6	19.1	21.9	23.0	22.7	21.1	17.1	12.6	9.5	(24)
(25)		10%	DB	7.2	7.6	12.7	18.6	23.9	27.4	28.1	27.4	25.6	20.4	15.1	9.0	(25)
(26)			MCWB	4.4	4.4	9.0	13.2	18.1	21.2	22.5	22.0	20.4	15.8	11.0	6.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.8	14.6	16.3	19.2	22.9	24.6	26.2	25.2	24.2	21.4	17.2	15.0	(27)
(28)			MCD B	16.7	18.9	22.4	25.2	28.6	29.7	30.7	29.8	28.8	26.3	20.1	17.5	(28)
(29)		2%	WB	11.3	11.2	14.4	17.4	21.4	23.6	25.0	24.2	23.1	19.8	15.1	12.3	(29)
(30)			MCD B	13.5	14.1	19.2	23.2	26.9	28.7	29.5	28.3	27.2	23.9	18.4	14.8	(30)
(31)		5%	WB	7.9	7.5	11.8	15.4	20.1	22.8	24.1	23.5	22.0	18.0	13.3	9.4	(31)
(32)			MCD B	10.1	10.3	15.8	20.8	25.0	27.8	28.0	27.4	25.9	21.3	16.3	11.7	(32)
(33)		10%	WB	4.5	4.5	9.1	13.4	18.9	21.8	23.4	22.8	21.0	16.4	11.5	6.8	(33)
(34)			MCD B	7.2	7.5	13.0	18.0	23.4	26.4	27.3	26.5	24.8	19.9	14.6	8.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.4	6.8	7.2	8.1	7.8	7.1	6.6	6.6	7.1	6.9	6.5	5.6	(35)
(36)			MCD BR	10.5	12.0	13.5	13.6	10.9	9.6	8.4	8.4	9.2	9.9	9.8	9.1	(36)
(37)			MCWBR	9.0	9.4	9.1	8.2	6.5	5.2	4.9	4.8	5.3	6.0	7.1	7.4	(37)
(38)		5% WB	MCD BR	10.4	11.6	12.7	12.9	9.9	8.2	6.8	7.1	7.7	8.3	8.9	9.0	(38)
(39)			MCWBR	9.4	9.7	9.1	8.6	6.1	4.8	4.7	4.6	5.0	5.7	7.2	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.308	0.337	0.361	0.400	0.436	0.454	0.454	0.438	0.390	0.371	0.329	0.311	(40)	
(41)		taud	2.398	2.273	2.288	2.269	2.202	2.196	2.221	2.253	2.362	2.392	2.483	2.473	(41)	
(42)		Ebn at Noon	840	860	883	871	844	826	822	824	843	815	811	809	(42)	
(43)		Edh at Noon	81	107	118	130	142	143	139	130	109	94	75	70	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.36	2.23	3.26	4.46	5.37	5.96	5.98	5.28	4.20	2.67	1.68	1.10	(44)	
(45)		RadStd	0.12	0.28	0.33	0.39	0.46	0.44	0.38	0.24	0.38	0.23	0.27	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page